

**Improving Operational Efficiency in B2C
Agribusiness: A Data-Driven Analysis for Sairanga
Pesticides & Seeds**

A Final Submission for the BDM Capstone Project

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1. EXECUTIVE SUMMARY

This Capstone Project on Business Data Management (BDM) examines sales and inventory performance at Sairanga Pesticides & Seeds, a B2C agribusiness outlet located in Kanaganapalli, Sri Sathya Sai District, Andhra Pradesh. The study analyzes operational data collected between August 2024 and July 2025 to evaluate product-wise profitability, customer purchasing behaviors, stock efficiency, and demand trends. The project aims to support informed decision-making and improve business sustainability in a competitive rural market.

A major insight from the analysis was the absence of structured tracking mechanisms, resulting in inefficiencies in inventory planning and profitability assessment. Using ABC classification, high-value items such as frequently purchased fertilizers were categorized as priority products for timely replenishment, while slow-moving crop protection products were identified for controlled stocking to prevent capital lock-up.

Customer segmentation using RFM analysis revealed distinct purchasing patterns, highlighting high-engagement customers who contributed significantly to revenue, along with irregular buyers who require targeted engagement strategies.

Time series forecasting was performed using a SARIMA model, which demonstrated strong alignment with actual sales patterns and provided reliable monthly demand predictions to support proactive inventory decisions.

The findings conclude with recommendations including data driven stock replenishment, customer focused sales strategies, and continued predictive forecasting to improve operational efficiency, minimize waste, and support sustainable business growth.

Overall, the analysis shows that applying data driven strategies in inventory control, pricing, and customer engagement can significantly improve efficiency and profitability. By prioritizing key products, forecasting seasonal demand, and focusing on high-value customers, the business can operate more strategically and support sustainable growth.

2. DETAILED EXPLANATION OF ANALYSIS/METHOD

2.1 Overview of the Analysis Process

The purpose of this analysis is to evaluate the operational performance of Sairanga Pesticides & Seeds, with a focus on product demand trends, customer purchasing patterns, inventory efficiency, and profitability. The study aims to address key business challenges such as irregular sales cycles, inventory buildup in low-demand products, lack of customer segmentation, and limited visibility into profit performance at a product level.

The analysis utilizes a primary dataset recorded from August 2024 to July 2025, containing monthly sales records, inventory movement, product pricing, category information, and customer purchase history. This data enabled an in-depth assessment of seasonal fluctuations, stock levels, revenue patterns, and product performance.

A structured analytical methodology was applied, beginning with manual data extraction and cleaning, followed by exploratory data analysis, descriptive statistics, and advanced analytical techniques including ABC product classification, RFM customer segmentation, profitability modeling, and forecasting using both SARIMA model. Visualizations were generated to interpret patterns and support decision-making.

This analytical framework enabled the identification of insights to optimize stocking decisions, strengthen customer engagement strategies, and improve product-wise profitability, ultimately supporting the growth and operational efficiency of Sairanga Pesticides & Seeds.

Kindly refer the folder for related documents:[BDM_DATA](#)

2.2 Data Collection and Preprocessing

The dataset for this analysis was recorded manually in Excel at the retail outlet during day-to-day business operations between August 2024 and July 2025. The data included essential variables such as Product ID, Product Name, Category, Cost Price, Selling Price, Units Sold, Revenue, Profit, and Customer Name. Inventory-related details including Opening Stock, Incoming Stock, and Closing Stock were documented monthly for each product to track availability and movement.

Python libraries such as pandas and Matplotlib were used for data cleaning and preprocessing and the following steps were executed:

Loading the Data:The analysis utilized two primary datasets imported into Pandas DataFrames. The sales dataset contained 2733 records across 10 columns documenting customer-wise product transactions, while the inventory dataset included 40 product entries with detailed month-wise stock tracking.

Handling Missing Values : Missing entries in critical fields such as cost price and selling price were corrected by referencing the official product rate sheet maintained by the shop. All updates were verified manually using physical invoice records to ensure accuracy and prevent data inconsistencies.

Data Cleaning and Standardization : Data standardization involves correcting spelling variations in product names (e.g., "V Green" standardized to "V-Green"), formatting all dates to DD-MM-YYYY for uniform temporal analysis, and standardizing measurement units and product category labels. This ensured consistency across all analytical dimensions.

Removing Duplicates :Duplicate sales entries were identified and removed using Pandas drop_duplicates() method. This step was essential to maintain dataset integrity and avoid inflated revenue or quantity metrics.

Product Categorization : Products were systematically grouped into meaningful divisions including Biostimulants, Fertilizers, Fungicides, Herbicides, Insecticides, Micronutrients, and Seeds. This categorization supported structured insights into product-level demand and profitability.

Inventory and Sales Data Structuring : Inventory information was organized into three temporal fields: Opening Stock, Incoming Stock (purchases), and Closing Stock. This structure provided clear visibility into monthly stock flow and enabled forecast-based stocking decisions.

These preprocessing steps ensured dataset quality and reliability, forming a strong foundation for subsequent analysis including ABC classification, RFM segmentation and forecasting using SARIMA models.

2.3 MetaData and Descriptive Statistical Analysis

Following the preprocessing phase, the finalized dataset consisted of structured records containing customer-wise sales details, product information, and inventory movements from August 2024 to July 2025. The sales dataset captured essential attributes such as product category, quantity sold, pricing, revenue, profit values, and customer names, while the inventory dataset included monthly stock movement fields such as opening stock, incoming stock, and closing stock for each product.

Descriptive analysis revealed noticeable seasonal sales patterns, with March 2025 recording the highest revenue, aligning with peak crop preparation activities, while January 2025 showed the lowest sales, reflecting reduced agricultural activity after the harvest period. Among all product categories, Biostimulants and Fertilizers contributed the highest share of sales and revenue, whereas Micronutrients and Seeds accounted for the lowest contribution. The monthly revenue trend showed a stable median range, indicating consistent customer purchasing behavior, though occasional bulk transactions led to revenue spikes during farming-intensive months.

Inventory movement analysis demonstrated clear alignment with agricultural cycles, where incoming stock increased prior to high-demand seasons. However, certain products displayed overstocking patterns during early 2025, suggesting lower-than-expected movement. Product-level insights highlighted Gronior, VN PLUS, and ACT as top-performing products, while items such as Formula 6 and Di-Ammonia Phosphate consistently ranked among the least sold. These findings, supported by descriptive statistical measures and visual analysis, provided a strong foundation for advanced analysis including demand forecasting, customer segmentation, and profitability evaluation, enabling data-driven decision-making for stock planning and business growth.

2.4 Customer Segmentation Analysis (RFM Model)

To understand customer purchasing behaviour and identify loyalty-driven revenue opportunities, an RFM (Recency, Frequency, Monetary) analysis was performed. The dataset was grouped customer-wise using `pandas.groupby()`, allowing structured analysis of purchasing patterns at the individual customer level.

For each customer, three behavioural indicators were computed:

$$Recency_i = T_{current} - T_{last_purchase,i}$$

$$Frequency_i = |\text{Number of purchases made by customer } i|$$

$$Monetary_i = \sum_{j=1}^{n_i} Revenue_{ij}$$

Where:

$T_{current}$ = most recent transaction date recorded

n_i = total number of purchases made by customer i

$Revenue_{ij}$ = monetary value of purchase j made by customer i

Next, each metric was transformed into a quantile-based score (1 = lowest, 5 = highest) using `pandas.qcut()` to standardize the distribution. A combined RFM score was then derived as

$$RFM_Score_i = R_Score_i + F_Score_i + M_Score_i$$

Based on RFM score thresholds, customers were segmented into Platinum, Gold, Growth Potential, Occasional, At-Risk, and Dormant groups.

The proportion of customers in each segment was calculated using

$$SegmentShare_k = \frac{N_k}{\sum_{j=1}^6 N_j} \times 100$$

Additionally, contribution to total revenue from each segment was computed as

$$Revenue_k = \sum_{i \in k} Monetary_i$$

These metrics enabled identification of high-value customers and segments requiring retention strategies such as promotional offers, reminders, or loyalty programs.

Tools Used

- RFM scoring using `pandas.qcut()`
- Segmentation using conditional logic
- Visualization through Pie Chart and Bar Chart using `matplotlib`

The customer segmentation analysis using the RFM model provided clear differentiation among purchasing behaviors, helping uncover patterns in loyalty, purchase frequency, and revenue contribution.

The segmentation results, visualized through pie and bar charts, revealed distinct customer tiers ranging from high value Platinum and Gold customers to low-engagement At Risk and Dormant groups. This structured framework enables the business to design targeted interventions such as customized offers for loyal buyers, re-engagement strategies for infrequent customers, and retention campaigns for declining segments. By leveraging a quantile-based scoring approach and automated Python workflows, the methodology ensures objective and scalable segmentation. Overall, this analysis provides a data driven foundation for improving customer relationship management, optimizing marketing efforts, and strengthening long-term business growth.

2.5 Product Demand Insights and Forecasting (SARIMA Model)

To improve sales prediction accuracy and support inventory planning, a Seasonal SARIMA model was applied to the monthly sales data from Sairanga Pesticides & Seeds. The analysis was performed using Python's statsmodels library and followed a systematic time-series modeling workflow.

$$SARIMA(p, d, q)(P, D, Q)_s$$

Where:

p: AR order (1)

d: Differencing (1)

q: MA order (1)

P: Seasonal AR (1)

D: Seasonal differencing (1)

Q: Seasonal MA (1)

s: Seasonality (12 months)

For the sales data, the following preprocessing steps were performed:

- Stationarity was achieved through first-order differencing (d=1) and seasonal differencing (D=1).
- Autocorrelation (ACF) and Partial Autocorrelation (PACF) plots were used to identify suitable autoregressive and moving-average orders.
- The presence of strong seasonal patterns following the agricultural cycle confirmed the need for a 12-month seasonal component.

These diagnostics guided the selection of the final SARIMA configuration:

- Non-seasonal component (1,1,1): modeled trend patterns and short-term fluctuations
- Seasonal component (1,1,1,12): captured annual sales seasonality and repeating demand behaviour

Tools Used

- Stationarity Testing: Augmented Dickey Fuller (ADF) Test
- Parameter Optimization: AIC based model tuning
- Model Fitting: Maximum Likelihood Estimation (MLE)
- Visualization & Diagnostics: Matplotlib Time-Series Plots

The SARIMA model produced accurate in-sample forecasts using `.get_prediction()` and generated a reliable three-month forward sales forecast using `.get_forecast(steps=3)`. The forecast closely followed observed historical patterns, indicating strong model stability and suitability for real-world planning. This enabled proactive inventory decisions, minimizing the risk of stock shortages during peak farming months and avoiding excess inventory during low-demand periods. Overall, the model strengthened the demand forecasting process for Sairanga Pesticides & Seeds and supports scalable, data-driven operational planning.

2.6 ABC Classification (Quantity Based Demand Segmentation)

To categorize products based on their total annual demand, an ABC classification was performed using the total units sold for each product. The process began by aggregating sales using

$$Q_i = \sum_{t=1}^n UnitsSold_{i,t}$$

Where:

Q_i = total quantity sold for product i

t = month of sale

n = total number of periods analyzed

Next, products were sorted in descending order of Q_i , and a cumulative percentage contribution was computed

$$CumPct_i = \frac{\sum_{k=1}^i Q_k}{\sum_{j=1}^m Q_j} \times 100$$

Where:

m = total number of products

k = product index after sorting

Based on the cumulative percentage thresholds, the products were classified using the following rule

$$Class = \begin{cases} A, & \text{if } CumPct_i \leq 70\% \\ B, & \text{if } 70\% < CumPct_i \leq 90\% \\ C, & \text{if } CumPct_i > 90\% \end{cases}$$

The resulting classification groups the products into three demand-based categories:

- Category A (High Demand – Green): Top ~70% of total units sold
- Category B (Moderate Demand – Orange): Next ~20% of sales volume
- Category C (Low Demand – Red): Final ~10% of sales volume

Tools Used

- Data Processing: Pandas
- Segmentation Logic: Conditional Thresholding
- Classification Automation: Python Function Mapping
- Visualization: Matplotlib

The ABC classification was performed using automated quantity-based segmentation and visualized to clearly distinguish fast, medium, and slow moving products. This analysis uncovered meaningful demand variations, supporting targeted stocking decisions and reducing the risk of over- or under-inventory. The systematic approach, built on cumulative sales distribution and Pareto-style categorization, ensured objective prioritization of high movement items. Python tools were utilized to streamline computation and visualization, making the process efficient and scalable for large product datasets.

2.7 Profitability and Scenario Modeling

To understand which products generate the highest financial impact, profitability was calculated at both the unit and total product level using the following formulas

$$UnitProfit_j = SP_j - CP_j$$

$$TotalProfit_j = UnitsSold_j \times (SP_j - CP_j)$$

Category-wise contribution to overall profitability was then derived by aggregating product-level profits within each category

$$ProfitCategory_c = \sum_{j \in c} TotalProfit_j$$

To estimate the potential effect of pricing strategy adjustments, a scenario simulation was conducted in which the selling price was increased by **10%** and the cost price was reduced by **10%**. The revised pricing values were calculated as

$$SP_{j,new} = 1.10 \times SP_j$$

$$CP_{j,new} = 0.90 \times CP_j$$

Using these updated price values, the adjusted profitability was computed as

$$Profit_{j,new} = UnitsSold_j \times (SP_{j,new} - CP_{j,new})$$

Finally, the improvement in profitability resulting from the scenario was determined using

$$\Delta Profit_j = Profit_{j,new} - TotalProfit_j$$

Tools Used:

- Vectorized calculations using Pandas
- Scenario comparison using conditional logic
- Visualization using comparative stacked bar charts

This analysis supports informed pricing and product strategy decisions by identifying which products contribute most to profitability at both unit and category levels.

The scenario simulation provides insight into how pricing adjustments such as increasing selling price or reducing cost can improve margins. By applying this structured approach, businesses can refine pricing policies, prioritize high impact products, and enhance overall revenue efficiency while reducing the risk associated with trial-and-error decisions.

2.8 Visualisations of Findings

Visualisation played a key role in interpreting the analytical outputs and supporting data driven insights throughout the study. Pie charts and bar charts were used to represent RFM based customer segmentation, clearly illustrating the distribution of customer groups and their contribution to revenue. Time-series line plots were used to compare historical sales patterns with SARIMA forecasts, helping validate seasonal behaviour and assess prediction accuracy. For product level insights, bar charts and pie charts highlighted category wise and product-wise profitability distributions, while scenario comparison charts displayed the impact of pricing adjustments on unit profit. ABC Classification results were visualised using colour-coded bar charts, enabling easy identification of fast moving, moderate-moving, and slow-moving products to support stocking strategies. Across all analyses, visualisation ensured clarity, pattern recognition, and effective communication of insights. All charts were generated using Python libraries such as Matplotlib and Seaborn, ensuring consistency, reproducibility, and accuracy throughout the analysis.

3. RESULTS AND FINDINGS

3.1 Customer Segmentation Analysis

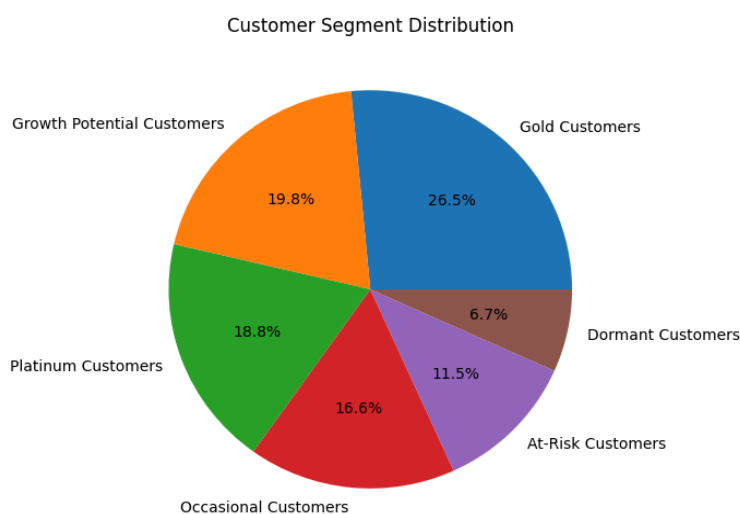


Fig:3.1.1:Customer Segment Distribution

Key Insights:

The Customer Segment Distribution Chart (Fig: 3.1.1) presents a structured overview of purchasing behavior across Sairanga Pesticides & Seeds' customer base, enabling a clearer understanding of loyalty strength, engagement patterns, and revenue risks. This visualization is essential for addressing the key challenges highlighted in the problem statement, particularly the need for improved customer retention, targeted marketing, and data-driven decision-making.

Gold Customers represent the largest segment, indicating a strong base of consistent and high-value buyers who demonstrate trust and repeated purchasing patterns. Growth Potential Customers and Platinum Customers also constitute a significant share and show promising engagement, suggesting opportunities for focused follow-ups, loyalty rewards, and personalized product recommendations to convert them into high-value lifetime customers.

Occasional Customers contribute moderately, yet their irregular buying patterns imply seasonal dependency or limited product awareness. This segment can be optimized through reminder-based selling and promotional strategies. Meanwhile, At Risk and Dormant Customers indicate declining or minimal engagement, signaling underlying gaps such as unmet expectations, switching behavior, or lack of follow up.

These insights emphasize the need for retention focused outreach, loyalty reinforcement, and strategic re-engagement to maximize long-term profitability and strengthen overall customer lifecycle value.

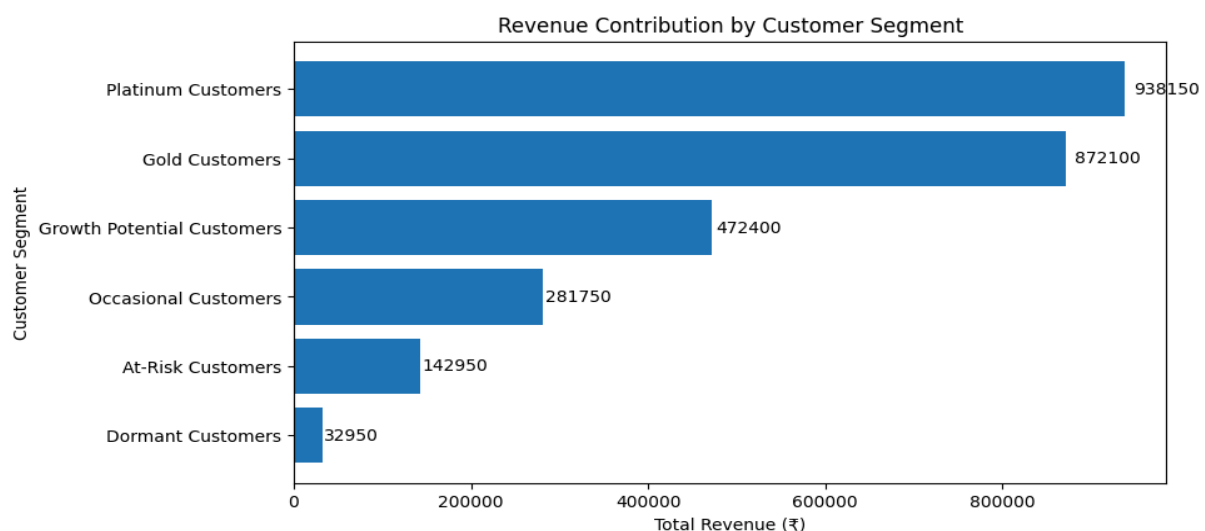


Fig:3.1.2:Revenue Contribution by Customer Segment

Key Insights :

The Revenue Contribution by Customer Segment chart (Fig: 3.1.2) provides a performance-based comparison of how different customer groups contribute to the overall sales of Sairanga Pesticides & Seeds. This visualization plays a key role in supporting the business problem statement by identifying revenue dependence, customer value tiers, and opportunities for strategic engagement to maximize profitability.

Platinum Customers contribute the highest revenue, showing strong purchase consistency and high-value transactions, making them the most profitable segment. Gold Customers follow, displaying similar loyalty-driven behavior and emphasizing the need to maintain strong relationships with these top-tier groups. Growth Potential Customers contribute a moderate share, indicating promising engagement that can be further developed through targeted communication, seasonal offers, and product bundling strategies.

Occasional Customers contribute a smaller but notable share of revenue, indicating irregular purchase patterns driven by short-term needs or limited product awareness. At-Risk and Dormant Customers contribute the least, reflecting reduced activity and potential churn due to limited follow-up, pricing concerns, or competitive options.

Overall, this analysis underscores the need to protect and nurture high-value segments while implementing reactivation strategies and loyalty-building programs for lower-engagement groups to strengthen retention and ensure sustainable revenue growth.

3.2 Product-Level Demand Insights (Volume Based ABC Segmentation)

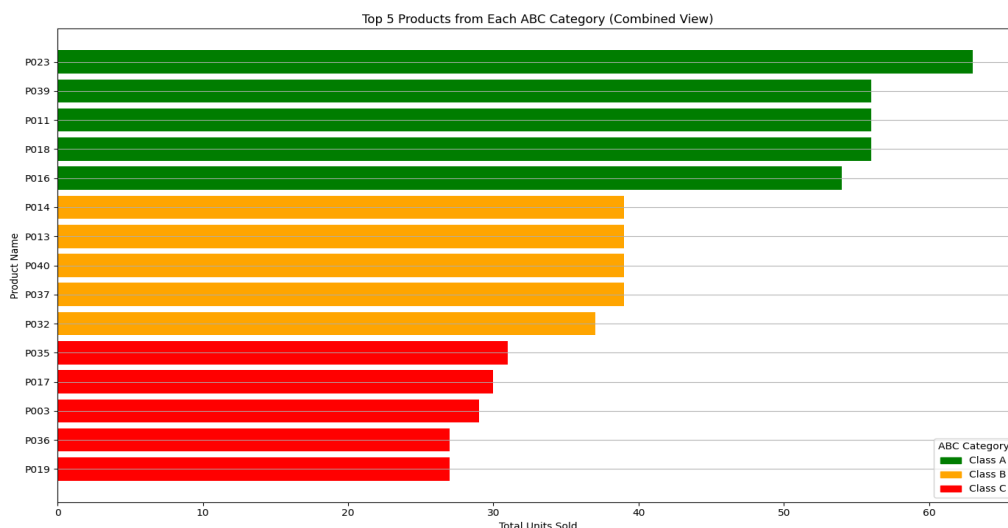


Fig:3.2.1:ABC Classification

Key Insights :

The ABC Classification Chart (Fig: 3.2.1) offers a structured segmentation of products based on their total annual demand, using aggregated sales volume as the core metric. This analysis directly aligns with the problem statement by helping identify critical inventory priorities and optimize purchasing decisions based on actual market movement rather than assumption-based stocking. After calculating total units sold and ranking products by cumulative contribution to demand, three product categories emerged, reflecting clear behavioral patterns in farmer purchasing trends.

Class A products, including P023, P039, P011, P018, and P016, fall within the highest cumulative sales range and represent the most in-demand items, requiring consistent replenishment planning, safety stock allocation, and prioritized vendor coordination to avoid stockouts during peak agricultural cycles. Class B products show moderate yet steady movement, indicating meaningful demand that can be enhanced through strategies such as bundling, advisory selling, or farmer awareness programs, with potential to elevate select items into Class A over time. In contrast, Class C products contribute the lowest share of sales and may lead to overstocking or increased storage costs, making them suitable for repositioning through promotional pricing, seasonal demand mapping, or controlled supply adjustments.

Overall, the classification highlights the need for differentiated inventory management strategies to improve operational efficiency and support data-driven stock planning.

3.3 Monthly Sales: Actual vs. Predicted

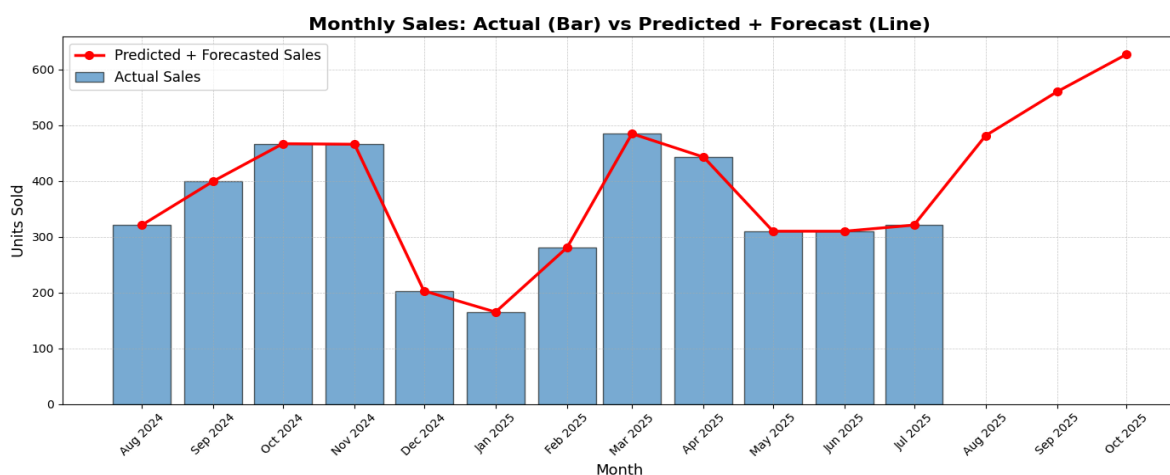


Fig:3.2.2: Monthly Sales: Actual vs. Predicted

Key Insights :

The Monthly Sales: Actual vs Predicted Forecast Chart (Fig: 3.2.1) highlights clear seasonal demand patterns and demonstrates how forecasting can support more accurate inventory planning. Actual sales show a recurring trend where demand increases between September and November, decreases during December and January, and rises again from February to April due to crop preparation and protection cycles. The SARIMA forecast closely follows these variations, validating its suitability for predicting product demand and helping address the current challenge of inconsistent stocking and unplanned purchasing.

Further, the model predicts a steady increase from August 2025 onward, peaking in October 2025. This signals the need for proactive ordering and vendor coordination to maintain adequate stock during upcoming high-demand periods. The comparison between observed and predicted values emphasizes the benefits of data-driven decision-making for procurement strategy, reducing the risk of stockouts during peak months and preventing excess holding costs during low-demand periods. Overall, the analysis reinforces that forecasting can support operational efficiency and ensure smoother demand supply alignment for Sairanga Pesticides & Seeds.

3.4 Product Wise Margin Analysis for Profitability Improvement

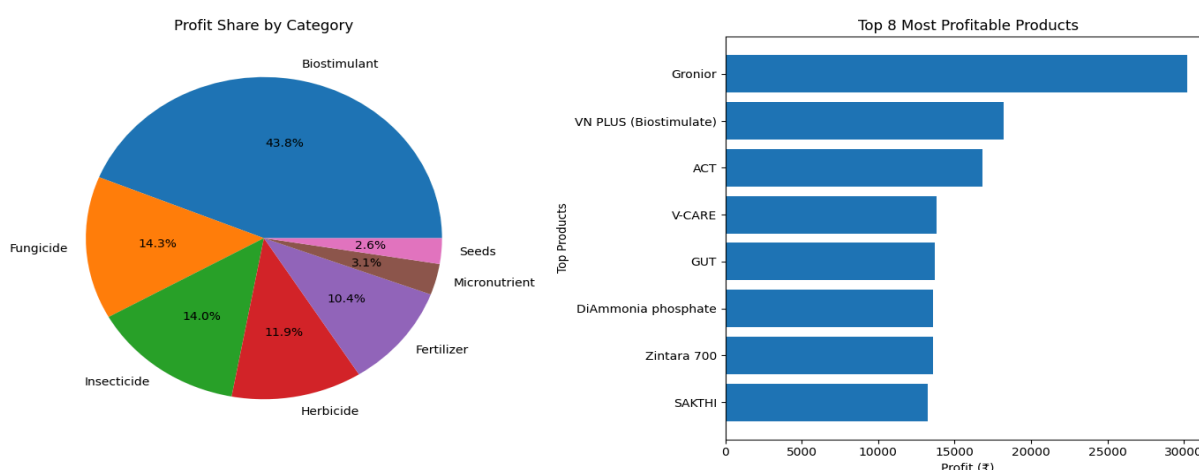


Fig:3.4.1: Category-wise contribution to total profit and Most profitable products based on gross margin

Key Insights :

The profit distribution chart shows that Biostimulants generate the highest share of overall profit, followed by Fungicides and Insecticides, indicating these categories play a major role in business earnings. Fertilizers and Herbicides contribute moderately, while Micronutrients and

Seeds add very little profit, suggesting limited demand or lower pricing margins. This uneven profit contribution highlights the need to prioritize stocking and promotion of high-performing product categories

The top profitable products chart further confirms that only a few items, such as Gronior, VN PLUS, and ACT, drive the majority of profit. Other products like V-CARE, Di-Ammonia Phosphate, and Zintara 700 contribute steady returns but at a lower scale. These insights suggest maintaining strong supply for top-performing products while reviewing pricing or sales strategies for low-profit categories to improve overall profitability.

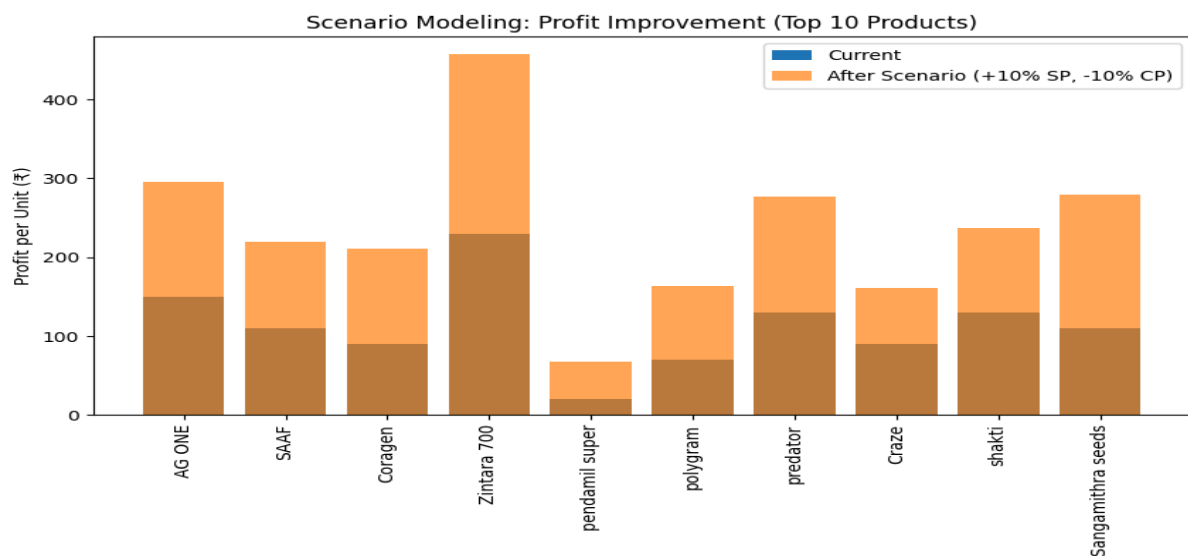


Fig:3.4.2:Scenario Modeling

Key Insights :

The Scenario Modeling chart compares the current profit per unit with the updated profit after applying a 10% increase in selling price and a 10% decrease in cost price for the top 10 products. The chart clearly shows a significant improvement in profit margin across all items under the new scenario. Products such as Zintara 700, AG ONE, and Predator show the highest increase, indicating that pricing adjustments have the greatest impact on already high demand and high value products.

The results also show that the improvement is not uniform across all products. While some items experience a large profit jump, others like Pendamil Super and Coragen show only moderate improvement. This suggests that low performing products may need additional strategies such as supplier negotiations, increased product awareness, bundled offers, or inventory control, rather than relying solely on pricing changes.

Overall, this scenario analysis supports the project objective of improving operational efficiency through data-driven decision making. The results show that even small selling price adjustments combined with cost optimization can significantly increase profitability without depending on higher sales volume. This helps the business identify where strategic price changes can deliver the greatest benefit, especially for top-performing products.

4. INTERPRETATIONS OF RESULTS AND RECOMMENDATIONS

4.1 Enhancing Customer Retention Through Segmentation Analysis

4.1.1 Interpretation

The customer segmentation analysis helps understand which customer groups contribute more to sales and which require more focus. Platinum and Gold customers are the most valuable, as they buy frequently and generate the highest revenue. These customers show strong trust and consistent purchasing habits, making them key to business growth. Growth Potential customers also show good buying behavior and can become loyal customers with proper follow up.

Occasional, At Risk, and Dormant customers contribute very little to sales. Their buying pattern is irregular, which may be due to lack of follow up, seasonal demand, or reduced need. This highlights the need for better engagement and communication strategies, especially before farming seasons, to increase their activity and prevent losing them permanently.

4.1.2 Recommendations

To improve customer retention, high value groups such as Platinum and Gold should be prioritized through loyalty benefits, personalized offers, and timely communication. Automated reminders aligned with farming cycles, crop based product suggestions, and seasonal promotions can help maintain consistent purchasing and strengthen loyalty. For low engagement groups such as Occasional, At Risk and Dormant customers, reactivation strategies are important, which may include sending discount messages, offering bundled deals, and conducting product awareness sessions to rebuild interest. Maintaining a communication schedule can help keep contact consistent and prevent customers from becoming inactive.

A simple rewards or points based system can encourage repeat purchases, while using past purchase history to suggest suitable fertilizers, pesticides, or micronutrients at the right time creates a more personalized experience.

Regularly tracking customer behavior through a CRM will also help identify early signs of reduced engagement and allow timely follow up.

4.2 Optimizing Product Level Sales Through Demand Analysis

4.2.1 Interpretation

The ABC Classification analysis helps identify which products contribute most to overall sales and which need better planning. Category A products such as P023 and P039 are high demand items that generate a large share of total sales. These products are essential for the shop and must be restocked regularly to avoid missed selling opportunities.

Category B products show moderate and steady demand, meaning they should be stocked based on monthly movement trends and seasonal patterns rather than in bulk and Category C products move slowly and take a longer time to sell, and stocking them in large quantities can lead to excess inventory and unnecessary stock holding costs.

The SARIMA forecasting model shows a close match between predicted and actual sales trends throughout the year, confirming that the model is reliable. Seasonal patterns are clearly visible, especially during crop preparation months, which highlights the importance of planning inventory based on seasonal demand cycles.

4.2.2 Recommendations

To improve product-level planning and prevent both shortages and excess stock, inventory decisions should be guided by demand patterns and ABC classification insights. Category A products should be monitored closely and restocked frequently, as they contribute the most to sales and have consistent movement throughout the year. Maintaining safety stock levels for these products will help avoid lost sales during peak agricultural seasons.

Category B items should be stocked in controlled quantities and reviewed periodically based on monthly demand trends. Running occasional promotional offers or farmer awareness campaigns may help increase movement and improve profitability for these products.

For Category C items, stocking should remain conservative to avoid holding unnecessary inventory. These products may also benefit from alternative strategies such as bundling, low-cost promotions, or supplier negotiation to improve turnover.

4.3 Improving Profitability Through Scenario Based Pricing Strategy

4.3.1 Interpretation

The profitability analysis shows that only a few products contribute significantly to total profit, especially high-performing items such as Gronior, VN PLUS, and ACT. These products have strong demand and better margins, making them important revenue drivers. The scenario modeling, where the selling price was increased by 10% and the cost price was reduced by 10%, demonstrates a clear improvement in profit levels. Products like Zintara 700, Predator, and AG ONE show the highest response, indicating that small pricing adjustments can create noticeable gains in overall profitability.

However, not all products showed the same level of improvement. Some items had only a small change in profit even after adjusting the price. For these products, pricing alone may not be enough, and additional strategies such as supplier negotiations, awareness programs, or bundling offers may be required. This indicates that a flexible approach rather than a uniform pricing change is more effective for profit improvement.

4.3.2 Recommendations

To improve profitability, pricing adjustments should be applied selectively, focusing on products that respond positively to price changes. High demand and high margin items should be prioritized for price optimization, while products with lower responsiveness should be reviewed for supplier cost reductions or promotional strategies. Introducing bundled offers, seasonal pricing, or loyalty based discounts can further motivate purchase frequency without affecting profit margins negatively.

In addition, monitoring product level margins regularly will help identify when adjustments are required. Supplier negotiations for bulk deals, early payments, or long term partnerships may help reduce the cost of low margin items. By combining pricing strategy, supplier coordination, and promotional planning, the business can improve overall profitability without relying solely on higher sales volume. This approach will support sustained financial growth and more efficient pricing decisions.